

SCHOOL MANAGEMENT SYSTEM

Product Requirements Document & MVP Development Plan

Dual-Curriculum (GES/NaCCA + ABEKA) | Ghanaian Private Basic School

Version 1.0 | June 2026

DRAFT — FOR INTERNAL USE

1. EXECUTIVE SUMMARY

1.1 What This System Does

This School Management System (SMS) is a web-based platform built specifically for a private basic school in Ghana that operates from Creche/Nursery through Junior High School (JHS 3 / Basic 9). The school runs two curricula simultaneously: the GES/NaCCA Standards-Based Curriculum (the national Ghanaian curriculum) and the ABEKA curriculum (a structured, American faith-based programme).

The system will handle every major administrative and academic function the school needs day to day: enrolling and tracking students, managing classes and subjects, recording attendance, entering and computing grades, generating report cards, billing parents, recording payments, and communicating with families. It will also help the school maintain its NaSIA operating licence and comply with Ghana's Data Protection Act, 2012 (Act 843).

1.2 Business Goal

Replace manual, paper-based or spreadsheet-based school administration with a single, integrated platform that:

- Accurately supports both GES/NaCCA and ABEKA grading and reporting in one system
- Reduces administrative burden on teachers and office staff
- Gives school leadership real-time visibility into academic performance, attendance, and finances
- Enables parents to stay informed without requiring phone calls to the school
- Positions the school for NaSIA compliance and future digital growth

1.3 Source Document Caveats

The research document used as input for this PRD contains useful technical research but also several weaknesses that must be acknowledged before development begins:

Research Document Limitations

Several sources cited are from third-party school websites, homeschool blogs, or consultancy privacy policies rather than official GES/NaCCA/ABEKA primary sources. The ABEKA GPA conversion tables, SBA weightings, and NaSIA compliance requirements should be verified directly with ABEKA Academy, GES district offices, and NaSIA before implementation. The data protection analysis draws partly from an EdTech company's own privacy policy rather than the official DPC text. Treat all specific figures (percentages, weightings, timelines) as indicative pending school-level confirmation.

2. PRODUCT VISION

2.1 Vision Statement

A practical, locally grounded school management platform that makes running a dual-curriculum Ghanaian private school simpler, more transparent, and more compliant — without requiring technical staff or expensive infrastructure.

2.2 Design Principles

- Simple over clever: favour clear screens over powerful but complex ones
- Ghana-first: Mobile Money, GES terminology, Ghanaian calendar, pesewa arithmetic
- Dual-curriculum without double work: one student record drives both GES and ABEKA outputs
- Progressive rollout: an MVP that works well, then grows
- Offline-tolerant: key functions (attendance, grades) must work on poor connections
- Privacy by design: Act 843 compliance built in from day one, not bolted on later

2.3 Guiding Scope Decisions

This PRD deliberately excludes from MVP: USSD portals, WhatsApp Business API, mobile apps, library management, transport, inventory, payroll, and AI-generated report comments. These are real features but would delay a usable first release by 12–18 months. The MVP must be deployable within 6–9 months and usable on day one of a new academic term.

3. USER ROLES & PERMISSIONS

3.1 Role Summary

Role	Primary Responsibility	Key Restrictions
Super Admin	System owner; manages all settings, users, backups, and configuration	Should not be a day-to-day school user; only for technical/owner access
School Administrator	Manages all school operations, users, and data	Cannot override audit logs
Headteacher / Principal	Academic leadership; approves report cards; views all academic data	Cannot delete payment records
Academic Coordinator	Manages curriculum, timetable, subjects, assessment types, and report templates	Cannot access financial records
Class Teacher	Manages own class: attendance, scores, report card comments for assigned class only	Cannot access other classes or financial data
Subject Teacher	Enters scores for assigned subjects only; views class roster	Cannot access student personal data beyond name and class
Accountant / Bursar	Manages all fees, billing, payments, and financial reports	Cannot access academic grades or report cards
Admissions / Reception	Manages student enquiries, admissions, and guardian records	Cannot access financial totals or academic grades
Parent / Guardian	Views own child's attendance, grades, report cards, fee balance, and notices	Read-only; cannot edit data; children's data only
Compliance Officer	Views NaSIA compliance reports, staff licensing logs, and audit trails	Read-only across compliance-relevant data
Student	Optional in MVP: views own timetable and published results only	Highly restricted; MVP may defer this role

3.2 Permission Matrix (Key Actions)

Action	Super Admin	Admin	Head	AcadC coord	ClassTeacher	SubjectTeacher	Bursar	Admissions	Parent
Enrol student	✓	✓	✓		✓			✓	
Edit student record	✓	✓	✓					✓	
Manage users/roles	✓	✓							

Action	SuperAdmin	Admin	Head	AcadCoord	ClassTeacher	SubjectTeacher	Bursar	Admissions	Parent
Create academic year/term	✓	✓	✓	✓					
Assign teachers to class	✓	✓	✓	✓					
Enter attendance	✓	✓	✓		✓				
Enter assessment scores	✓	✓	✓		✓	✓			
Generate report cards	✓	✓	✓	✓					
Approve/publish report cards	✓	✓	✓						
Create fee invoices	✓	✓					✓		
Record payments	✓	✓					✓		
View own child's data									✓
View all financial reports	✓	✓	✓				✓		
View compliance reports	✓	✓	✓						
Delete/archive records	✓	✓							
Configure grading formulas	✓	✓		✓					

4. CORE MODULES & REQUIREMENTS

4.1 Student Admissions & Records

- Admissions enquiry log (name, age, curriculum interest, contact)
- Student application form with: full name, DOB, gender, nationality, religion, medical alerts, previous school
- Guardian records: primary and secondary guardian, relationship, phone, email, occupation, emergency contact
- Ghana Card ID field (optional at admission; required for older students per GES)
- Student photo upload
- Admission status workflow: Enquiry → Application → Offered → Enrolled → Active
- Student unique ID auto-generated (format: school prefix + year + sequence, e.g. GPS-2026-0042)
- Sibling linkage (for sibling fee discounts)
- Transfer and withdrawal recording with reason codes
- Student archive on exit (records retained per Act 843 / GES policy; never deleted)

Verification Required

Ghana Card ID is referenced as a mandatory field in the research document. Confirm with school and GES district office whether this is required at admission or only at later stages. Do not make it a hard-required field in MVP — make it optional with a flag.

4.2 Class & Level Management

- Level structure: Creche, Nursery, KG1, KG2, Basic 1–9 (GES naming) mapped to ABEKA equivalents
- Streams/sections per level (e.g. Basic 4A, Basic 4B)
- Class capacity settings
- Class teacher assignment per class per term
- Subject teacher assignments per subject per class per term
- Student-to-class enrollment per academic year
- Promotion workflow (end of year): bulk promote, retain, or graduate

4.3 Curriculum Management

- Two curriculum tracks configured at school level: GES_NACCA and ABEKA
- Each student enrollment tagged to a primary curriculum track
- For hybrid/dual students: enrollment can link to both tracks (MVP: flag as 'dual'; V2: full parallel gradebook)
- Level-to-curriculum mapping (which levels use which curriculum or both)

Critical Design Decision for MVP

The research document proposes full parallel gradebooks for dual-curriculum students. This is technically correct but complex for MVP. Recommendation: In MVP, each student is assigned ONE primary curriculum track. The system supports switching between GES and ABEKA report templates. True dual-track parallel assessment (same student, two separate gradebooks simultaneously) should be deferred to Version 2 unless the school confirms this is a day-one requirement.

4.4 Subject Management

- Subject catalogue: name, code, curriculum track, level applicability
- GES/NaCCA subjects: English, Mathematics, Science, Social Studies, RME, French, History, ICT, OWOP, Creative Arts, Physical Education (per NaCCA curriculum framework)
- ABEKA subjects configured separately (may overlap in name but differ in assessment structure)
- Subject assignment to classes per term
- Credit hours/weight per subject (for ABEKA GPA calculation)

Confirm with School

The exact subject list, especially for early childhood levels (Creche, Nursery, KG), must be confirmed with the school. The NaCCA KG curriculum uses activity-based learning strands, not formal subject names. The school may use its own subject naming convention.

4.5 Academic Calendar & Term Management

- Academic year creation (e.g. 2026/2027)
- GES term structure: 3 terms per year; configurable start/end dates
- ABEKA quarter structure: 4 quarters per year, 180-day instructional calendar
- Term/quarter status: Draft → Active → Closed
- Holiday and school event calendar per academic year
- Prevent duplicate active years/terms

Simplification for MVP

In MVP, support the GES 3-term calendar as the primary calendar. Add ABEKA quarters as a secondary calendar layer. Full parallel scheduling (running both on separate tracks simultaneously) is a V2 feature. Most practical dual-curriculum schools in Ghana run ABEKA within the GES term structure.

4.6 Attendance

- Daily attendance per class: Present, Absent, Late, Excused
- Attendance recorded by class teacher

- Attendance summary per student per term (days present, days absent, attendance %)
- Bulk attendance entry (mark all present, then flag absences)
- Late/absent SMS notification to parent (MVP: manual trigger; V2: automatic)
- Attendance report per class, per level, per school
- Lock attendance after N days to prevent backdating (configurable)

4.7 Assessment & Grading

This is the most complex module. See Section 9 (Grading Model) for full detail. Summary requirements:

- Configurable assessment types per curriculum track and level (Class Exercise, Quiz, Project, SBA, Mid-Term Exam, End-of-Term Exam, Oral Reading, etc.)
- Score entry interface: teacher enters raw marks per assessment per student
- GES/NaCCA grading engine: SBA + Exam weighted formula; profile dimensions for JHS
- ABEKA grading engine: percentage-to-letter-grade mapping; GPA calculation
- Grade scale configuration (admin-configurable, not hardcoded)
- Academic eligibility flag for ABEKA (flag students below 76% average)
- Lock scores after report card publication

4.8 Report Cards

- Report card templates configurable per curriculum track and per level group
- GES/NaCCA template: subject scores, performance level descriptors, SBA/exam breakdown, attendance summary, teacher and headteacher remarks
- ABEKA template: subject scores, letter grade, GPA, attendance, teacher and headteacher remarks
- Early childhood (Creche–KG): developmental milestone report (not numerical grades)
- Draft → Review → Approved → Published workflow
- PDF generation per student
- Bulk PDF generation per class
- Print-ready format

Confirm with School

Report card format is the single most important thing to confirm with the school before building this module. The school may have an existing format they want replicated exactly, or they may have GES district-specific requirements. Build the template engine flexibly, but do not design the actual template until you have seen the school's current report card.

4.9 Fees & Billing

- Fee categories: Tuition, Development Levy, Feeding, Books/Materials, Uniform, Examination, Miscellaneous
- Fee schedule per level per term (different amounts for different classes)

- Invoice generation per student per term (auto-generated when term opens, or manual)
- Part payments: invoice tracks total due, total paid, outstanding balance
- Arrears carried forward from previous term
- Sibling discount configuration (e.g. 10% off second child)
- Scholarship/bursary recording (full or partial fee waiver with reason)
- Manual payment recording: cash, bank deposit, cheque (receipt auto-generated)
- All monetary values stored in pesewas (GHS × 100) to avoid floating-point errors
- Receipt generation (printable PDF)
- Debtor report: list of students with outstanding balances
- Fee reminder notices (manual send to parent contact)

4.10 Payment Integration (Mobile Money)

- Integration with ONE payment gateway in MVP (recommended: Paystack or Hubtel — confirm with school's bank)
- Parent pays via MoMo link sent to their phone
- Webhook handler with HMAC-SHA512 signature verification
- Idempotency check: duplicate webhook events must not create duplicate payment records
- Server-side transaction verification before marking payment as successful
- Never trust client-side redirect parameters for payment confirmation
- Bank settlement period: T+1 for Paystack/Hubtel

KYC / Onboarding Action Required

Before going live with any payment gateway, the school must complete KYC onboarding: Certificate of Incorporation, GRA TIN, corporate bank account, Ghana Card of proprietor/directors, and verified physical address (Ghana Post GPS). This takes time — start this process early, in parallel with development.

4.11 Parent Communication

- Broadcast announcements: send to all parents, a specific class, or individual parent
- Fee reminders with outstanding balance
- Report card availability notification
- Attendance absence alert
- SMS as primary channel in MVP (via bulk SMS gateway: e.g. Arkesel, mNotify, or Hubtel SMS)
- Notification log per student/guardian
- Parent portal: web-based login to view child's attendance, grades, report card, and fee balance

WhatsApp & USSD: Version 2

The research document recommends WhatsApp Business API and USSD. Both are valuable for Ghanaian parents but add significant complexity (WhatsApp requires Meta Business verification; USSD requires a telco short-code agreement). Both are deferred to Version 2. MVP uses SMS + parent web portal.

4.12 Staff & Teacher Records

- Staff profile: name, photo, contact, role, department, employment type
- NTC (National Teaching Council) registration number and licence status
- Police clearance record and expiry date
- Medical certificate record and expiry date
- Professional development log
- Subject and class assignments (linked to academic module)
- Staff status: Active, On Leave, Resigned, Terminated

4.13 NaSIA / GES Administrative Reports

- Pupil-to-teacher ratio report per level
- Pupil-to-classroom ratio report
- Staff NTC licensing status report (licensed, induction, unlicensed)
- Staff police clearance and medical certificate status with expiry flags
- Enrolment statistics by level, gender, curriculum track
- Infrastructure/asset registry (science lab, library, ICT lab, sports area, first aid)

NaSIA Compliance Complexity

The research document implies a full automated NaSIA compliance dashboard. This is aspirational for MVP. In MVP, build the data foundation (staff records, ratio data, asset registry) and provide exportable reports. A full compliance workflow dashboard (tracking Part A/Part B licensing steps) is a Version 2 feature.

4.14 Data Protection & Privacy (Act 843)

- Parental consent recorded at enrollment (digital confirmation logged with timestamp and IP)
- Granular consent: educational processing vs. marketing/third-party use
- Children's data flagged as Special Personal Data (all students under 18)
- Data encrypted in transit (TLS 1.3) and at rest (AES-256)
- Role-based access control enforced at API level, not just UI level
- Audit log for all data access, modification, and deletion events
- Data retention schedule configured and enforced (see Section 11)
- Parent self-service: request data export, request correction

- Data breach response workflow: internal incident flag → notification payload to DPC within 72 hours



Legal Verification Required

The research document's data protection analysis is based partly on third-party EdTech company privacy policies, not the full text of Act 843. Before development, obtain and read Act 843 directly, or engage a Ghanaian data protection lawyer to review the compliance checklist (Section 11 of this PRD). The school must also register with the Data Protection Commission (DPC) as a Data Controller.

4.15 Settings & Configuration

- School profile: name, logo, address, motto, Ghana Post GPS, phone, email
- Academic structure configuration (levels, streams, curriculum tracks)
- Grading scale configuration per curriculum and level group
- Assessment type configuration per level/curriculum
- Fee category configuration
- Report card template configuration
- SMS/notification gateway configuration
- Payment gateway configuration (API keys, webhook secrets)
- Data retention policy settings

4.16 User Management & Permissions

- User account creation (email + password; SMS OTP for password reset)
- Role assignment (one or more roles per user)
- Account activation / deactivation
- Password policy (minimum length, forced reset on first login)
- Session timeout configuration
- Two-factor authentication (optional in MVP; recommended for Admin and Bursar roles)

4.17 Audit Logs & Backups

- Immutable audit log: who did what, to which record, at what time, from which IP
- Audit log covers: login events, data creation, edits, deletions, exports, payment events
- Audit log cannot be edited or deleted (even by Super Admin)
- Automated daily database backups to off-site/cloud storage
- Backup retention: 30 days rolling
- Manual backup trigger available to Super Admin
- Restore procedure documented (not necessarily in-app in MVP)

5. MVP SCOPE

The MVP must be deployable in 6–9 months and usable from the first day of an academic term. The following is a strict scope definition. Resist the temptation to add features — every addition delays the launch date.

5.1 Must-Have (MVP Core)

Module	Must-Have Features
Student Records	Enrolment, guardian records, student profiles, transfer/withdrawal, student ID generation
Class Management	Level/stream setup, class teacher assignment, student-to-class placement
Curriculum Setup	Two tracks (GES, ABEKA), level-to-track mapping, curriculum-specific subjects
Academic Calendar	Academic year + GES 3-term calendar creation and management
Attendance	Daily attendance per class, term summary, basic attendance report
Assessment & Grades	Score entry per assessment type, automated grade calculation (GES + ABEKA), grade display
Report Cards	GES report card template (primary + JHS), ABEKA report card template, PDF export, approval workflow
Early Childhood Reports	Developmental milestone report template for Creche–KG
Fees & Billing	Fee schedule, invoice generation, part payments, arrears, manual payment recording, receipts, debtor report
MoMo Payment	Single gateway integration (Paystack or Hubtel), webhook + idempotency, server-side verification
Parent Communication	SMS broadcast, fee reminders, report card notification, absence alert (manual send)
Parent Portal	Web login, view child's attendance, grades, report cards, fee balance
Staff Records	Staff profile, NTC number, police clearance, medical certificate expiry flags
User Management	Roles, permissions, user creation, password reset
Basic Admin Reports	Enrolment counts, gender breakdown, attendance summary, debtor list, pupil-teacher ratio
Settings	School profile, grading scale config, assessment type config, fee category config
Data Protection	Consent logging at enrolment, TLS encryption, role-based access, basic audit log
Backups	Automated daily backup, manual backup trigger

5.2 Should-Have (MVP if Time Allows)

- Sibling fee discount automation
- Scholarship/bursary recording
- Bulk promotion wizard (end of year)
- NaSIA-ready infrastructure/asset registry
- Two-factor authentication for Admin/Bursar
- Configurable report card remarks bank (pre-written teacher comment options)

5.3 Nice-to-Have (Include Only if Zero Scope Risk)

- Student photo upload in profile
- CSV/Excel import for bulk student enrolment
- Calendar view of school events and term dates
- Subject teacher score-entry portal (separate from class teacher view)

5.4 Explicitly Deferred to Version 2

Feature	Reason for Deferral
WhatsApp Business API	Requires Meta Business verification, WhatsApp template approval — 4–6 week process minimum
USSD Portal	Requires telco short-code agreement, separate USSD application server — major separate project
Mobile App (iOS/Android)	Not needed for MVP; parent portal works on mobile browser
Full dual-track parallel gradebook	Complex schema; confirm whether school actually needs same student in two gradebooks simultaneously
NaSIA compliance dashboard	Data foundation in MVP; workflow dashboard in V2
AI-generated report card comments	Gimmick without data history; defer until system has 1–2 years of data
Library management	Separate system or V2 module
Transport management	Separate system or V2 module
Staff payroll	Typically handled in separate accounting software; confirm with school
Advanced analytics / dashboards	Build basic reports first; analytics layer after 1–2 terms of data
Parent data export self-service portal	Log consent and access requests manually in MVP; automate in V2
ABEKA Academy transcript upload	Confirm whether school actually submits to ABEKA Academy for validation
Inventory management	Out of scope for MVP

Feature	Reason for Deferral
Multi-school / multi-branch support	Single school only for MVP

6. VERSION 2 ROADMAP

Version 2 development should begin after at least one full academic term of live MVP operation, using real user feedback to prioritise.

6.1 Priority 1: Communication Upgrade

- WhatsApp Business API: rich-media report cards, consent forms, announcements
- Automated absence alerts (triggered by attendance save, not manual)
- Automated fee reminders (configurable schedule: 7 days before due, on due date, 7 days overdue)
- Parent notification preferences (SMS vs WhatsApp vs email)

6.2 Priority 2: Full Dual-Track Gradebook

- Parallel assessment records for students enrolled in both GES and ABEKA simultaneously
- Separate GES and ABEKA report cards for the same student in the same term
- ABEKA Academy transcript format export
- ABEKA 180-day instructional day tracking (separate from GES term calendar)

6.3 Priority 3: NaSIA Compliance Dashboard

- Interactive NaSIA inspection readiness checklist
- Part A and Part B licensing workflow tracker
- Automated staff credential renewal reminders
- Pupil-to-teacher and pupil-to-classroom ratio health indicators
- Exportable NaSIA inspection data package

6.4 Priority 4: USSD Portal

- Basic parent USSD menu (balance check, grade summary, consent confirmation)
- Requires telco short-code agreement — start negotiation in parallel with V2 development

6.5 Priority 5: Advanced Analytics

- Class and subject performance trends over time
- At-risk student identification (falling grades, attendance below threshold)
- Fee collection velocity and revenue forecasting
- Teacher performance summary (attendance of their class, grade distribution)

6.6 Priority 6: Operational Modules

- Library management: book catalogue, issue/return, overdue tracking
- Transport management: routes, vehicle assignment, parent confirmation

- Basic staff payroll (or integration with existing payroll software)
- Inventory / stationery tracking

6.7 Priority 7: Mobile App

- Progressive Web App (PWA) first — works on mobile browser, installable
- Native iOS/Android only if PWA adoption is insufficient
- Parent-focused: attendance alerts, grade notifications, fee balance, report card download

7. KEY WORKFLOWS

7.1 New Student Admission

1. Admissions officer creates an Enquiry record (student name, guardian contact, level of interest, curriculum preference)
2. Guardian submits application form (online or paper-to-system entry)
3. School reviews application; status updated to Offered or Rejected
4. On acceptance: admissions officer creates Student Profile; system generates unique Student ID
5. Guardian record created/linked; consent form presented and logged (with timestamp and IP)
6. Student assigned to a class and curriculum track for the current academic year
7. Fee invoice generated for the relevant term
8. Welcome SMS/notification sent to guardian

7.2 Student Class Placement

9. Academic coordinator opens Class Placement screen for an active academic year/term
10. Selects a class; student appears in 'Unenrolled' list if admitted but not yet placed
11. Drags/assigns student to class (or uses bulk CSV import)
12. System validates: no duplicate enrollment in same academic year for same curriculum track
13. Class teacher and subject teachers automatically gain access to the student in their gradebook

7.3 End-of-Year Promotion

14. Academic coordinator opens Promotion wizard at end of year
15. System lists all students per class with their annual performance summary
16. Coordinator marks each student as: Promoted, Retained, Graduated (JHS 3), Withdrawn
17. Graduated students (JHS 3 / Basic 9) are archived with full academic record
18. Promoted students are pre-assigned to the next class for the new academic year (pending confirmation)
19. New academic year is created; term dates configured
20. Fee schedules updated for new year; invoices generated when term opens

7.4 Creating an Academic Year & Term

21. Super Admin or Admin creates a new Academic Year (e.g. 2027/2028)
22. Configures three terms with start date, end date, and exam dates
23. Sets term status to Draft

- 24. Reviews class assignments and fee schedules for the new year
- 25. Activates Term 1: status changes to Active; attendance and score entry opens
- 26. At term end: Admin closes term (locks attendance and scores); triggers report card workflow

7.5 Taking Attendance

- 27. Class teacher logs in on any device (desktop or mobile browser)
- 28. Selects date (defaults to today) and class
- 29. System shows full class list; all students default to Present
- 30. Teacher marks individual students as Absent, Late, or Excused
- 31. Teacher submits; attendance record locked after N days (configurable, e.g. 3 days)
- 32. Absent students: system flags for optional manual SMS alert to guardian

7.6 Entering Assessment Scores

- 33. Teacher (class or subject) navigates to gradebook for their assigned class/subject
- 34. Selects assessment type (e.g. Class Exercise 1, Mid-Term Test, End-of-Term Exam)
- 35. Enters raw marks per student (system shows max mark for that assessment type)
- 36. System auto-calculates weighted grades based on configured formula
- 37. Teacher reviews and saves; scores are visible but not yet published to parents
- 38. Scores lock after report card is approved

7.7 Generating Report Cards

- 39. Academic coordinator triggers report card generation for a class/level after term closes
- 40. System computes final grades, performance levels, GPA (ABEKA), and attendance summary
- 41. Class teacher adds conduct rating and teacher remarks (free text or from remarks bank)
- 42. Headteacher reviews and adds headteacher remarks; approves report card batch
- 43. On approval, system publishes report cards: parents can view in portal; PDF available for download/print
- 44. Optional: SMS notification sent to guardians that report cards are available

7.8 Creating Fee Bills & Recording Payments

- 45. Bursar opens fee schedule for the term; confirms or adjusts fee items per level
- 46. System generates invoices for all active enrolled students (or bursar generates individually)
- 47. Bursar or parent views invoice showing: itemised fees, total due, any arrears, total outstanding

- 48. Manual payment: Bursar records cash/cheque payment → system updates outstanding balance → PDF receipt generated
- 49. MoMo payment: Bursar sends payment link to parent phone → parent pays via MoMo → webhook confirms → balance updated → SMS receipt sent
- 50. Bursar can view debtor report at any time: students with outstanding balances

7.9 Sending Parent Communication

- 51. Admin or class teacher navigates to Communication module
- 52. Selects audience: All Parents / Specific Class / Individual Guardian
- 53. Selects message type: Announcement / Fee Reminder / Report Card Notification / Absence Alert
- 54. Composes message (or selects template)
- 55. Previews recipient count and message content
- 56. Sends; system logs: sent time, recipient count, message content, sender
- 57. Failures (invalid numbers) are flagged for follow-up

7.10 Exporting Administrative Reports

- 58. Admin or compliance officer navigates to Reports module
- 59. Selects report type: Enrolment Stats / Attendance / Finance / NaSIA / Staff Compliance
- 60. Filters by academic year, term, level, curriculum track as needed
- 61. System generates report in-browser; download as PDF or Excel
- 62. Report generation is logged in the audit trail

8. DATA MODEL

The following describes the main entities, their purpose, and key fields. This is a product-level model, not full SQL DDL. Use this to design your database schema and API contracts.

8.1 Core Identity Entities

School

Single-school MVP. Stores the school's profile.

- id, name, logo_url, address, ghana_post_gps, phone, email, motto, registration_number, created_at

AcademicYear

Represents a school year (e.g. 2026/2027). Parent of Terms.

- id, school_id, label (e.g. '2026/2027'), start_date, end_date, is_active, created_at

Term

A term within an academic year. GES uses 3 terms; ABEKA uses 4 quarters (V2).

- id, academic_year_id, term_number (1/2/3), label, start_date, end_date, exam_start_date, status (draft/active/closed), curriculum_track (GES/ABEKA/both)

Level

An educational level (e.g. KG1, Basic 4). Maps to both GES and ABEKA designations.

- id, school_id, name (e.g. 'Basic 4'), ges_designation, abeka_designation, order_index, level_group (ECE/Lower Primary/Upper Primary/JHS), is_active

ClassRoom

A specific class section (e.g. Basic 4A). Child of Level.

- id, level_id, section_label (e.g. 'A'), capacity, academic_year_id, class_teacher_id (FK: Staff), is_active

8.2 Student & Guardian Entities

Student

Core student identity record. Persists across academic years.

- id (UUID), student_number (generated, e.g. GPS-2026-0042), first_name, middle_name, last_name, preferred_name, date_of_birth, gender, nationality, religion, ghana_card_id (nullable), profile_photo_url, previous_school, medical_alerts (text, encrypted), admission_date, status (active/transferred/graduated/withdrawn), created_at, updated_at

Guardian

A parent or guardian. One student may have multiple guardians.

- id, first_name, last_name, relationship (Father/Mother/Uncle/Guardian/etc.), phone_primary, phone_secondary (nullable), email (nullable), occupation, is_primary_contact, consent_granted_at, consent_ip_address, consent_details (JSONB), created_at

StudentGuardian

Junction table linking students to guardians (many-to-many).

- id, student_id, guardian_id, is_primary

Enrollment

Links a student to a specific class and curriculum track for a given academic year.

- id, student_id, classroom_id, academic_year_id, curriculum_track (GES_NACCA/ABEKA), enrollment_date, status (active/transferred/withdrawn/completed), exit_date (nullable), exit_reason (nullable)

8.3 Staff Entities

Staff

All school employees. Used for teacher assignment and NaSIA compliance.

- id, first_name, last_name, phone, email, staff_number, role_category (teacher/admin/support), ntc_registration_number (nullable), ntc_status (licensed/induction/unlicensed/not_applicable), police_clearance_date, police_clearance_expiry, medical_cert_date, medical_cert_expiry, employment_type (full_time/part_time/contract), status (active/on_leave/resigned), created_at

SubjectAssignment

Links a staff member to a subject in a specific class for a term.

- id, staff_id, subject_id, classroom_id, term_id

8.4 Curriculum & Subject Entities

CurriculumTrack

Configuration record for each curriculum the school supports.

- id, school_id, code (GES_NACCA / ABEKA), name, is_active

Subject

A subject in the school's catalogue.

- id, school_id, name, code, curriculum_track_id, level_group_applicability (JSONB array), credit_hours (for ABEKA GPA), is_active

ClassSubject

Links subjects to a classroom for a specific term.

- id, classroom_id, subject_id, term_id, is_active

8.5 Assessment Entities

AssessmentType

Configuration for each assessment component (e.g. Class Exercise, SBA, End-of-Term Exam).

- id, school_id, name, curriculum_track_id, level_group, max_score, weight_percentage, assessment_category (formative/summative/exam), is_active

AssessmentScore

A single student's raw score for a single assessment instance.

- id, enrollment_id, subject_id, assessment_type_id, term_id, raw_score, max_score, recorded_by (staff_id), recorded_at, is_locked

GradingScale

Maps score ranges to grade descriptors. Configurable per curriculum and level group.

- id, school_id, curriculum_track_id, level_group, min_score, max_score, letter_grade, descriptor (e.g. 'Excellent'), gpa_value (for ABEKA), is_active

ReportCard

A published report card for a student for a term. Stores computed results.

- id, enrollment_id, term_id, curriculum_track_id, computed_grades (JSONB), overall_grade, gpa (nullable, ABEKA only), attendance_summary (JSONB), teacher_remarks, headteacher_remarks, conduct_rating, status (draft/approved/published), approved_by (staff_id), approved_at, published_at

8.6 Finance Entities

FeeItem

A configurable fee line item (e.g. Tuition, Feeding, Books).

- id, school_id, name, category, is_active

FeeSchedule

Amount charged for a fee item for a specific level in a specific term.

- id, fee_item_id, level_id, term_id, amount_pesewas, is_active

Invoice

A billing record for a student for a term.

- id, student_id, term_id, generated_at, total_amount_pesewas, total_paid_pesewas, outstanding_pesewas, arrears_pesewas, discount_pesewas, discount_reason, status (unpaid/partial/paid/waived)

InvoiceLineItem

Individual fee line on an invoice.

- id, invoice_id, fee_item_id, description, amount_pesewas

Payment

A payment record against an invoice.

- id, invoice_id, student_id, amount_pesewas, payment_method (cash/bank_deposit/cheque/mobile_money), payment_reference, gateway_reference (nullable), paid_at, recorded_by (staff_id), receipt_number, is_verified, notes

PaymentGatewayLog

Immutable log of all webhook events received from payment gateway.

- id, gateway_reference, event_type, payload (JSONB), received_at, signature_valid, processed, invoice_id (nullable)

8.7 Communication Entities

Notification

A message sent to one or more guardians.

- id, school_id, sender_id (staff), type (announcement/fee_reminder/report_ready/absence_alert), subject, body, audience (JSONB: class_ids, student_ids, or 'all'), channel (sms/portal), sent_at, recipient_count, failure_count

8.8 Security & Compliance Entities

User

System login account, linked to either a Staff or Guardian record.

- id, email, phone, password_hash, linked_entity_type (staff/guardian), linked_entity_id, is_active, last_login_at, created_at

UserRole

Assigns one or more roles to a user.

- id, user_id, role_name, granted_by, granted_at

AuditLog

Immutable record of all system events. Must never be editable or deletable.

- id, user_id, action (login/create/update/delete/export/payment_event/etc.), entity_type, entity_id, changes (JSONB: before/after diff), ip_address, user_agent, timestamp

ConsentRecord

Stores verifiable consent for each student's data processing.

- id, student_id, guardian_id, consent_type (educational_processing/marketing), granted (boolean), granted_at, ip_address, method (otp/signature/checkbox), revoked_at (nullable)

8.9 Key Relationships Summary

From	Relationship	To
AcademicYear	has many	Terms
Term	has many	ClassSubjects, AssessmentScores, Invoices
Level	has many	ClassRooms
ClassRoom	has many	Enrollments, SubjectAssignments
Student	has many	Enrollments, AssessmentScores, Invoices, Payments
Student	has many (via StudentGuardian)	Guardians
Enrollment	has many	AssessmentScores
Enrollment	has one per term	ReportCard
Invoice	has many	InvoiceLineItems, Payments
Staff	has one	User
Guardian	has one	User (parent portal login)

9. GRADING & REPORT CARD MODEL

9.1 Design Principle

The grading engine must be configuration-driven, not hardcoded. Grade formulas, weights, and scales are set by the academic coordinator in the Settings module. This allows the school to adjust them without a developer.

9.2 GES/NaCCA Grading (Basic 1 – Basic 6: Lower & Upper Primary)

- Assessment components: Class Exercises, Class Tests, Project/SBA, End-of-Term Exam
- Typical weighting structure (confirm with school): Continuous Assessment (SBA) 50%, End-of-Term Exam 50% — or school-specific variation
- Performance levels (not just a score): Exceeding Standards (80–100%), Meeting Standards (60–79%), Approaching Standards (40–59%), Developing (0–39%)
- Report card shows: raw marks, total score, performance level descriptor, teacher remarks, attendance summary

Verify SBA Weights

The research document cites 60:40 and 50:50 SBA:Exam ratios. The correct ratio depends on the specific subject, year group, and GES guidance for that academic year. Do not hardcode any ratio. Build a configuration screen. Confirm the school's current practice with the GES district office and the school's academic coordinator.

9.3 GES/NaCCA Grading (Basic 7 – Basic 9: JHS / Common Core Programme)

- Profile dimension weightings per subject (example for Mathematics): Knowledge & Understanding 30%, Application of Knowledge 40%, Attitudes/Values/Process Skills 30%
- Each SBA component maps to one or more profile dimensions
- End-of-term exam also mapped to profile dimensions
- Final grade = weighted average across all profile dimensions
- JHS report card includes: profile dimension scores, total SBA, exam score, final grade, performance level, teacher remarks, headteacher remarks, attendance summary

Profile Dimension Complexity

Profile dimension weightings may vary by subject. The Mathematics CCP curriculum specifies 30/40/30, but other subjects may differ. The system must allow per-subject dimension weight configuration. Confirm all subject-specific weightings with the school's JHS coordinator before building this module.

9.4 ABEKA Grading

- Scoring: 0–100 numerical scale
- Assessment types: Daily exercises, Quizzes, Bi-weekly tests, Oral reading evaluations, Semester/Quarter exams
- Grade-to-letter conversion: configurable (use ABEKA's standard scale as default; allow school to adjust)
- GPA calculation (unweighted): $GPA = \frac{\sum(\text{Grade Points} \times \text{Credits})}{\sum(\text{Credits})}$
- Academic eligibility flag: students with unrounded average below 76% are flagged
- ABEKA report card: subject scores, letter grade, GPA, cumulative GPA, attendance, teacher remarks, headteacher remarks
- GPA scale: support both ABEKA Choice 1 and Choice 2 (configurable per school)

Score Range	Letter Grade	GPA Choice 1	GPA Choice 2
97–100	A+	4.00	4.00
93–96	A	3.75	4.00
90–92	A-	3.50	4.00
87–89	B+	3.25	3.00
83–86	B	3.00	3.00
80–82	B-	2.75	3.00
77–79	C+	2.50	2.00
73–76	C	2.25	2.00
70–72	C-	2.00	2.00
67–69	D+	1.75	1.00
63–66	D	1.50	1.00
60–62	D-	1.25	1.00
0–59	F	0.00	0.00

ABEKA GPA Scale Verification Required

The GPA scale above comes from a third-party source (ABEKA homeschool transcript guide and an Indiana Christian Academy website), not from an official ABEKA Academy Ghana-specific

Score Range	Letter Grade	GPA Choice 1	GPA Choice 2
document. Before implementing, verify with ABEKA Academy directly or with the school's ABEKA coordinator which GPA scale they use.			

9.5 Early Childhood (Creche, Nursery, KG1, KG2)

- No numerical grades for these levels
- Developmental milestones tracked: physical development, social/emotional, language, cognitive, creative
- Rating per milestone: Not Yet Observed / Beginning / Developing / Achieved
- ABEKA KG: includes foundational literacy tracking (oral reading, writing readiness, numeracy)
- Report card is a narrative/milestone-based report, not a marks sheet

Confirm with School

The specific milestones and categories for each age group must be designed with the school's ECE coordinator. This is highly school-specific and cannot be assumed from general curriculum documents.

9.6 Report Card Configuration

- Template types: GES Primary, GES JHS (CCP), ABEKA, ECE Developmental
- Each template configures: header (school name, logo, term, student info), subject rows, grade columns, remarks fields, attendance summary block, conduct/attitude section, signature lines
- Conduct/attitude rating: configurable scale (e.g. Excellent/Very Good/Good/Needs Improvement)
- Teacher remarks: free text (max 300 characters) or selectable from remarks bank
- Headteacher remarks: free text
- Attendance summary: days present, days absent, attendance percentage (auto-calculated)
- Promotion status: printed on report card (e.g. 'Promoted to Basic 5')

9.7 Questions to Confirm Before Building Report Cards

63. What does your current GES report card look like? (Request a sample)

64. What does your current ABEKA report card look like? (Request a sample)
65. Do you have a single report card template or different templates for different levels?
66. What SBA:Exam weighting do you use for each level group?
67. Do you use profile dimensions for JHS? If so, do you have the subject-specific weightings?
68. What conduct/attitude categories do you use?
69. Do you produce a single report card or separate GES and ABEKA reports for dual-track students?
70. Do you currently submit anything to ABEKA Academy for validation? If so, what format?

10. PAYMENT & BILLING MODEL

10.1 Fee Structure

Category	Description	Level-Specific?
Tuition	Core academic fee per term	Yes (different per level)
Development Levy	School infrastructure fund	Possibly — confirm with school
Feeding / Canteen	School lunch programme	Sometimes flat rate
Books & Materials	Textbooks, exercise books, stationery	Yes
Uniform	School uniform supply charge	Typically once per year
Examination Fee	Internal/external exam administration	JHS levels only
Extra-Curricular	Sports, clubs, special activities	Optional / as applicable
Miscellaneous	Any other ad-hoc charges	As applicable

10.2 Invoice Logic

- Invoice generated per student per term, covering all applicable fee items for their level
- Invoice includes: itemised fees, any arrears from prior term, any sibling discount applied, scholarship/waiver applied, gross total, total paid, outstanding balance
- Part payments: each payment reduces outstanding balance; invoice remains Open until fully paid
- Arrears: at start of new term, any unpaid balance from prior term is automatically added as an arrears line on the new invoice
- Invoice cannot be deleted once created; only voided with a reason (audit trail)

10.3 Discounts & Waivers

- Sibling discount: configurable % or fixed amount for 2nd, 3rd+ children of the same family
- Scholarship: partial or full waiver of tuition (or any fee item); requires reason and approver
- Staff child discount: if applicable — configurable
- All discounts and waivers logged with approver and reason for audit purposes

10.4 Manual Payment Recording (MVP Core)

- Bursar selects student → opens outstanding invoice
- Enters: amount, date, method (Cash / Bank Deposit / Cheque / MoMo manual)

- Enters reference (bank deposit slip number, cheque number, etc.)
- System updates outstanding balance, records payment, generates receipt
- Receipt includes: school header, student name, class, term, amount, method, reference, bursar name, date, receipt number

10.5 Mobile Money Integration (MVP)

- Recommended gateway for MVP: Paystack or Hubtel (confirm with school's preferred bank for settlement)
- Flow: Bursar generates MoMo payment link for specific invoice → link sent to guardian phone via SMS → guardian pays → webhook received → server verifies → balance updated → SMS receipt to guardian
- All amounts handled internally in pesewas (GHS × 100); displayed in GHS with 2 decimal places
- HMAC-SHA512 webhook signature verification before any database update
- Idempotency: check gateway_reference exists before processing; skip if already processed
- Server-side verification: always call gateway's verify endpoint before marking paid; never trust redirect parameters
- Payment gateway log: every webhook event stored raw for reconciliation

10.6 Reconciliation & Reporting

- Daily collection summary: total cash, total MoMo, total bank deposits per day
- Outstanding debtors report: all students with unpaid or partially paid invoices
- Term revenue report: expected vs. collected per fee category
- Payment history per student: full payment ledger
- Export reports as PDF or Excel

10.7 MVP vs V2 Payment Features

Feature	MVP	Version 2
Manual cash/bank/cheque payment recording	✓	
Single MoMo gateway integration (Paystack/Hubtel)	✓	
Webhook + HMAC verification + idempotency	✓	
Invoice generation and part-payment tracking	✓	
Arrears carry-forward	✓	

Feature	MVP	Version 2
Sibling discount	✓	
Scholarship/waiver recording	✓	
Receipt PDF generation	✓	
Debtor report	✓	
Automated fee reminders (scheduled SMS)		✓
Multi-gateway support (Paystack + Hubtel + Flutterwave)		✓
Parent self-service MoMo payment (parent-initiated)		✓
Direct debit / recurring payment setup		✓
Advanced fee analytics and forecasting		✓
Integration with accounting software (e.g. QuickBooks)		✓

11. REPORTS

11.1 Academic Reports

Report	Purpose	Access	MVP?
Class Performance Summary	Average scores per subject per class for a term	Head, AcadCoord, ClassTeacher	Yes
Student Academic Record	Full academic history: all terms, all subjects, all grades	Admin, Head, Parent (own child)	Yes
Subject Grade Distribution	Grade breakdown (A/B/C/D/F) per subject per term	Head, AcadCoord	Yes
Report Card (student PDF)	Printable report card per student per term	All relevant roles + Parent	Yes
ABEKA GPA Summary	GPA per student for ABEKA-track classes	Head, AcadCoord	Yes
Academic Eligibility Flag Report	ABEKA students below 76% threshold	Head, AcadCoord, ClassTeacher	Yes
Annual Performance Trend	Student's grade trend across multiple terms/years	Admin, Head	V2
At-Risk Students Report	Students with declining performance or high absences	Head, AcadCoord	V2

11.2 Attendance Reports

Report	Purpose	Access	MVP?
Daily Attendance Sheet	Attendance for a class on a given day	ClassTeacher, Admin	Yes
Student Attendance Summary	Days present/absent/late per student per term	ClassTeacher, Head, Parent	Yes
Class Attendance Summary	Attendance % per class for a term	Head, AcadCoord	Yes
Chronic Absentee Report	Students below a threshold attendance % (configurable)	Head, Admin	Yes
School-Wide Attendance Summary	Overall attendance by level for a term	Head, Admin	Yes

11.3 Finance Reports

Report	Purpose	Access	MVP?
Outstanding Debtors	All students with unpaid or partial invoices	Bursar, Admin, Head	Yes
Daily Collection Summary	Payments received by method on a given day	Bursar, Admin	Yes
Term Revenue Report	Expected vs. collected per fee category per term	Bursar, Admin, Head	Yes
Student Payment History	Full payment ledger for a student	Bursar, Admin, Parent (own)	Yes
Arrears Report	Students carrying arrears from prior term(s)	Bursar, Admin	Yes
Scholarship/Discount Register	All fee waivers and discounts applied	Admin, Head	Yes
Annual Fee Collection Summary	Revenue across all terms in an academic year	Admin, Head	V2
Fee Collection Forecast	Expected collections based on enrolment and fee schedule	Admin, Head, Bursar	V2

11.4 Student Records Reports

Report	Purpose	Access	MVP?
Enrolment Statistics	Total enrolled by level, gender, curriculum track	Admin, Head	Yes
New Admissions Report	Students admitted in a given term/year	Admin, Head, Admissions	Yes
Transfers & Withdrawals	Students who transferred in or out	Admin, Head	Yes
Class List	Full list of students in a class with contact details	ClassTeacher, Admin	Yes

11.5 Staff Reports

Report	Purpose	Access	MVP?
Staff Directory	All active staff with roles and contact details	Admin, Head	Yes
NTC Licensing Status	Staff NTC registration and licence status	Head, Compliance	Yes
Credential Expiry Report	Police clearances and medical certs expiring within 60 days	Admin, Head, Compliance	Yes

Report	Purpose	Access	MVP?
Pupil-to-Teacher Ratio	Ratio by level (for NaSIA compliance)	Admin, Head, Compliance	Yes

11.6 NaSIA / Administrative Reports

Report	Purpose	Access	MVP?
Enrolment Statistics (NaSIA format)	Headcount by level and gender for GES/NaSIA submission	Admin, Head, Compliance	Yes
Pupil-Teacher Ratio Report	For NaSIA inspection readiness	Admin, Compliance	Yes
Infrastructure Asset Register	List of school facilities and equipment	Admin, Compliance	Yes
Staff Compliance Summary	NTC, police clearance, medical cert status for all staff	Admin, Head, Compliance	Yes
NaSIA Inspection Readiness Checklist	Interactive compliance self-assessment	Admin, Head	V2

11.7 Compliance / Security Reports

Report	Purpose	Access	MVP?
Audit Log Export	Full audit trail for a date range or user	SuperAdmin only	Yes
Login History	All login events with IP and device	SuperAdmin, Admin	Yes
Consent Register	All data consent records with timestamps	Admin, Compliance	Yes
Data Retention Flags	Records flagged for archival or deletion per retention schedule	SuperAdmin, Admin	Yes
Payment Gateway Log	All webhook events received from payment gateway	SuperAdmin, Admin, Bursar	Yes
Breach Incident Log	Security incident records	SuperAdmin, Admin	Yes

12. PRIVACY & COMPLIANCE CHECKLIST

Based on Ghana's Data Protection Act, 2012 (Act 843) and school data requirements. Items flagged with ⚠️ require legal verification with a qualified Ghanaian data protection practitioner before launch.

12.1 Registration & Governance

- School must register with the Data Protection Commission (DPC) as a Data Controller
- Developer/platform operator should register as a Data Processor
- Appoint a Data Protection Supervisor (can be an existing staff member trained for the role)
- Data Processing Agreement between school and developer must be in place before launch

⚠️ Legal Verification Required

Confirm registration requirements and procedures directly with the DPC (dataprotection.org.gh). The research document cites third-party sources for Act 843 requirements — not the official Act text.

12.2 Data Collection & Consent

- Collect only data that is necessary (data minimisation principle)
- Obtain explicit consent from parent/guardian before collecting any student data
- All students are minors; all student data is Special Personal Data under Act 843
- Consent record must include: guardian name, date, time, IP address, method (checkbox/OTP/signature), what was consented to
- Separate consent for: educational processing vs. communications vs. third-party sharing
- No marketing use of student or guardian data without explicit separate consent
- Consent must be freely given (not a condition of enrollment — though educational processing consent will be practically required)

12.3 Access Control

- Role-based access control enforced at the API layer, not just the UI
- Minimum necessary access per role (principle of least privilege)
- Parent portal: parent sees only their own child's data
- Teachers see only their assigned classes and subjects
- Strong password policy; session timeout after inactivity
- All admin-level actions require re-authentication (or 2FA in V2)

12.4 Technical Security

- HTTPS / TLS 1.3 for all data in transit
- AES-256 encryption for sensitive data at rest (medical alerts, guardian contact details, financial records)
- Database not publicly accessible; accessed only via application layer
- Payment API keys and gateway secrets stored in environment variables, never in code or database
- Webhook endpoint protected by HMAC-SHA512 signature verification
- No client-side payment status updates

12.5 Audit & Accountability

- Immutable audit log for all data access, modification, deletion, export, and payment events
- Audit log retained for duration of account + 2 years minimum
- Audit log cannot be edited or deleted by any role including Super Admin
- Regular log review procedure (monthly recommended)

12.6 Data Retention Schedule

Data Category	Retention Period	Action on Expiry
Student academic records	Permanent (GES archiving policy)	Locked archive — never deleted
Student attendance records	7 years post-exit	Anonymised and scrubbed
Financial / payment records	7 years	Archived for tax compliance; then deleted from active tables
User accounts & technical logs	Account duration + 2 years	Secure deletion
Support communications	3 years from resolution	Purged
Marketing consent records	Consent duration + 3 years	Deleted from notification roster
Guardian contact data (post-exit)	Until erasure requested or 7 years	Review and notify

Retention Periods Need Verification

The retention periods above are drawn from the research document's sources which include a third-party EdTech privacy policy, not official GES archiving policy or GRA tax guidance.

Data Category	Retention Period	Action on Expiry
Verify with: (a) GES district office for academic record retention, (b) GRA for financial record retention, (c) DPC for personal data retention under Act 843.		

12.7 Data Subject Rights

- Right of Access: parent can request export of all data held on their child (MVP: manual process via admin; V2: self-service)
- Right to Rectification: parent can request correction of inaccurate data (admin-mediated in MVP)
- Right to Erasure: parent can request deletion when student transfers out — but school must balance this against GES academic archiving obligations
- All data subject requests logged with date, nature of request, and action taken

12.8 Cloud Hosting & Data Localisation

Act 843 restricts cross-border transfer of Ghanaian personal data without adequate protections.

- If hosting on international cloud (AWS, Azure, GCP): ensure DPC-approved data processing agreements are in place
- Preferred: host in a region that offers a local Ghana or West Africa zone (AWS af-south-1 Cape Town is closest; no Ghana-specific AWS region as of 2026)
- Consider hybrid: sensitive personal data (student records, guardian data) on local Ghanaian hosting; payment routing via international gateway
- Document hosting choices and justification before launch

Cloud Hosting Requires Legal Review

The cross-border data transfer requirements are complex and evolving. Engage a Ghanaian data protection lawyer to review the hosting architecture before go-live.

12.9 Breach Response

- Define a breach: any unauthorised access to student, guardian, or financial data
- Breach detection: monitor for unusual login patterns, bulk data exports, payment anomalies
- On confirmed breach: notify DPC within 72 hours (Section 31, Act 843)

- Notification payload to DPC: nature of breach, estimated affected records, mitigation steps taken, Data Protection Supervisor contact
- Notify affected individuals promptly
- Document all breach incidents in Breach Incident Log

13. RISKS & ASSUMPTIONS

13.1 Risk Register

Risk	Likelihood	Impact	Mitigation
Curriculum requirements not confirmed before build	High	High	Run discovery workshop with school before development. Do not build assessment engine until SBA weights and report card formats are confirmed.
Report card format rejected by school after build	High	High	Show mockups and paper prototypes to school before building. Get sign-off on template design before coding.
Dual-track gradebook scope creep in MVP	High	High	Strictly limit MVP to one curriculum track per student. Defer true parallel gradebooks to V2.
Payment gateway KYC delays going live	Medium	High	Start KYC onboarding process at project kickoff, in parallel with development.
Ghana Data Protection Act non-compliance	Medium	High	Engage a Ghanaian data protection lawyer for a compliance review before launch.
Internet connectivity issues for staff	High	Medium	Ensure critical features (attendance, score entry) have graceful offline handling or mobile-optimised UI with low bandwidth requirements.
Staff adoption resistance	Medium	Medium	Involve class teachers in UAT. Provide simple training materials in plain language. Start with low-friction features (attendance, parent messaging).
ABEKA-specific implementation differs from research	High	Medium	Verify all ABEKA requirements directly with ABEKA Academy or the school's ABEKA coordinator. Do not trust third-party blog sources.
Over-engineering the MVP	High	Medium	Enforce MVP scope strictly. Any new feature request during MVP build goes to the backlog, not the current sprint.
Parent portal low adoption	Medium	Low	SMS remains the fallback channel. The portal is a convenience, not a dependency.
NaSIA compliance requirements change	Low	Medium	Build NaSIA reports as configurable exports, not hardcoded formats. Monitor NaSIA website for updates.

13.2 Key Assumptions

- The school runs one academic year at a time (not concurrent years — e.g. no trimester/semester hybrid)

- The school uses the GES 3-term structure as its primary calendar, even for ABEKA classes
- Most parents have a mobile phone capable of receiving SMS
- The school has a reliable internet connection for staff to use the admin portal
- The school will designate at least one person as a system administrator and one as academic coordinator
- The school's current ABEKA implementation is self-managed (i.e. they do not formally register students with ABEKA Academy for external validation — confirm this assumption)
- Fee structures are set per level, not per individual student (except for scholarships/waivers)
- The school uses GHS (Ghanaian Cedi) as its only billing currency
- One payment gateway is sufficient for MVP

14. OPEN QUESTIONS FOR THE SCHOOL

These questions must be answered before or during a discovery workshop with the school. Do not proceed with building key modules without answers to the high-priority items.

14.1 Curriculum & Assessment (HIGH PRIORITY)

71. Does every student follow ONE curriculum (either GES or ABEKA), or do some students do both simultaneously?
72. For classes that use ABEKA: do you also submit GES report cards to GES/district office? Or only ABEKA reports?
73. Do you currently send any student records to ABEKA Academy for external validation or transcripts?
74. What SBA-to-Exam weighting do you use for each level group? (e.g. 50:50 or 60:40?)
75. Do you use profile dimensions (Knowledge/Application/Attitudes) for JHS grading, and if so, do you have the subject-specific weightings?
76. Can you share a copy of your current GES report card template?
77. Can you share a copy of your current ABEKA report card template?
78. Do you have different report card designs for different level groups (e.g. primary vs JHS)?
79. What assessment types do you currently use? (List the actual names you use: e.g. Class Work, Home Work, Mid-Term Test, End-of-Term Exam, Project, etc.)
80. For early childhood levels (Creche–KG), what categories do you assess? Do you use numerical scores or milestone descriptors?

14.2 Classes & Levels (HIGH PRIORITY)

81. What is the exact list of levels you have in the school, and how are they named? (e.g. Creche, Nursery 1, Nursery 2, KG1, KG2, Basic 1... Basic 9)
82. How many streams/sections per level? (e.g. Is there a Basic 4A and 4B?)
83. Do different streams within the same level follow different curricula, or the same?

14.3 Report Cards & Remarks

84. Do teachers type their own remarks, or do you have a standard list they choose from?
85. Who approves report cards before they are printed — the class teacher, head teacher, or both?
86. Do you currently print report cards and give them to parents, or do you share them digitally?

14.4 Fees & Payments

87. What are your fee categories? (e.g. Tuition, Feeding, Building Levy, Books, Exam)

- 88. Do fees differ by level? (e.g. KG fees vs JHS fees)
- 89. Do you currently offer sibling discounts? If so, what is the structure?
- 90. Do you offer scholarships or bursaries? How are these currently managed?
- 91. How do parents currently pay? (Cash at school / bank deposit / mobile money?)
- 92. Which Mobile Money networks do most of your parents use? (MTN / Telecel / AT?)
- 93. Which bank does the school use for settlements? (This affects which gateway to recommend)
- 94. Do you currently carry forward unpaid fees from one term to the next as arrears?

14.5 Parent Communication

- 95. Do you currently send SMS messages to parents? If so, which SMS provider do you use?
- 96. What percentage of your parents do you estimate have smartphones vs basic phones?
- 97. Do parents currently use any digital tool to check their child's grades or fees? (e.g. another SMS, WhatsApp group, etc.)

14.6 Admissions & Records

- 98. What is your current admissions process? Do you keep a waiting list?
- 99. Do you currently record Ghana Card IDs for students or parents? Is this required by your GES district office?
- 100. Do you have an existing student numbering system you want to preserve?
- 101. How do you currently record student transfers in and transfers out?

14.7 Staff & Compliance

- 102. Do all your teachers have NTC registration? How do you currently track this?
- 103. Are all staff police-cleared? Do you track certificate expiry dates?
- 104. Has the school obtained its NaSIA operating licence? What stage of the process are you at?
- 105. Has the school registered with the Data Protection Commission (DPC)?

14.8 Current Pain Points

- 106. What is the single most painful thing about managing the school today that you want this system to fix?
- 107. What do you spend the most administrative time on that feels like it should be automated?
- 108. What do parents complain about most in terms of communication or information access?
- 109. How do you currently generate report cards? (Excel, manual, another system?)

110. Have you used a school management system before? If so, what did you use and why are you moving away from it?

15. RECOMMENDED NEXT STEPS

15.1 Immediate Actions (Before Development Starts)

#	Action	Owner	Timeline
1	Conduct a 3–4 hour discovery workshop with school: answer all open questions in Section 14	Product Manager + School Admin	Week 1–2
2	Collect sample report cards (current GES and ABEKA templates) from the school	Product Manager	Week 1
3	Confirm SBA weightings and assessment types with school's academic coordinator	Product Manager	Week 1–2
4	Start payment gateway KYC onboarding (Paystack or Hubtel — confirm choice with school)	School Owner	Week 1 (parallel)
5	Engage Ghanaian data protection lawyer to review compliance requirements under Act 843	Project Lead	Week 2–3
6	Confirm cloud hosting strategy and data localisation approach	Technical Lead	Week 2–3
7	Register school with DPC as Data Controller (if not already done)	School Owner	Week 2–4
8	Verify ABEKA GPA scale and assessment requirements with ABEKA coordinator	Product Manager	Week 2
9	Design and get school sign-off on report card mockups (paper/Figma) before coding	Designer	Week 3–5

15.2 Technical Architecture Recommendations

- Backend: Node.js or Python (Django/FastAPI) — both have good Paystack/Hubtel SDK support
- Database: PostgreSQL — supports JSONB for flexible schema fields (grading config, consent details, computed grades)
- All monetary values: store as INTEGER in pesewas throughout the database
- Multi-tenant readiness: even for single-school MVP, design with a school_id on every table to make future multi-school support feasible without a database migration
- Report card PDF generation: use a headless PDF library (Puppeteer, WeasyPrint, or similar) — HTML-to-PDF gives the most layout control for complex report card formats
- Authentication: JWT with short expiry + refresh tokens; bcrypt for password hashing
- API: REST for MVP; no need for GraphQL complexity at this stage

- Hosting: recommend starting with a reputable African cloud provider (e.g. AWS Lagos af-south region or Render.com) — discuss localisation trade-offs with lawyer
- Frontend: React or Next.js — works well on mobile browsers without needing a native app

15.3 MVP Build Sequence

Phase	Modules	Estimated Duration
Phase 1: Foundation	School setup, academic year/term, levels/classes, user management, staff records, curriculum tracks	3–4 weeks
Phase 2: Students	Student admissions, guardian records, consent logging, class enrollment, student portal (parent login)	3–4 weeks
Phase 3: Academics	Subject management, attendance, assessment score entry, GES grading engine, ABEKA grading engine	5–6 weeks
Phase 4: Report Cards	Report card templates (GES Primary, GES JHS, ABEKA, ECE), PDF generation, approval workflow	4–5 weeks
Phase 5: Finance	Fee schedule, invoice generation, manual payment recording, receipts, MoMo integration, debtor report	4–5 weeks
Phase 6: Communication	SMS notifications, parent portal pages (grades, fees, report cards)	2–3 weeks
Phase 7: Compliance & Reports	Audit logs, data retention flags, NaSIA reports, admin reports, consent management	3–4 weeks
Phase 8: UAT & Launch	User acceptance testing with school staff, bug fixes, data migration (if any), training	3–4 weeks

Total estimated MVP build time: approximately 27–35 weeks (7–9 months) with a small team of 2–3 developers. This is only achievable if discovery is complete, report card templates are approved, and scope is held firm.

FINAL NOTE

This PRD is a starting point, not a contract. The most important next step is the discovery workshop with the school. Many critical decisions — grading formulas, report card formats, fee structures — cannot be made without the school's direct input. Build this system collaboratively with the school, not in isolation.